Homework 5

Problem 1 (15pts)

Compute the Fourier transform of each of the following signals:

(a) $x[n] = (\frac{3}{7})^{-n} u[-n]$

(b) $x[n] = \sin[\frac{\pi n}{3}] \cos[\frac{\pi n}{6}]$

(c) $x[n]$ is represented in Figure 1.

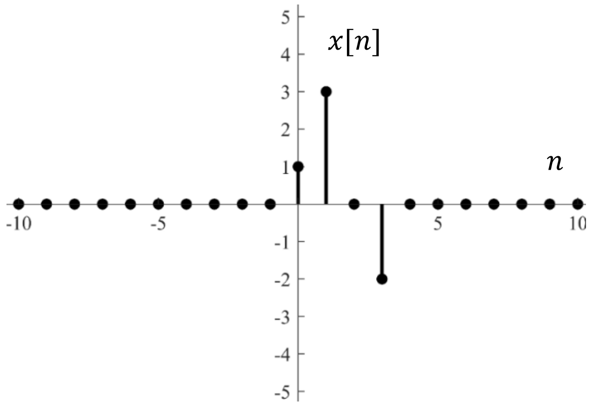

Figure 1: The discrete signal $x[n]$.

Problem 2 (10pts)

Compute the inverse DTFT of $X(e^{j\omega})$ of each of the following signals using the definition of the inverse DTFT:

(a) $X(e^{j\omega}) = \begin{cases} 1 & , \frac{\pi}{4} \leq |\omega| \leq \frac{3\pi}{4} \\ 0, & 0 \leq |\omega| \leq \frac{\pi}{4}, \frac{3\pi}{4} \leq |\omega| \leq \pi \end{cases}$

(b) $X(e^{j\omega}) = 2\sum_{k=-\infty}^{\infty}[\delta\left(\omega + \frac{2\pi}{5} + 2\pi k\right) + \delta\left(\omega - \frac{2\pi}{5} + 2\pi k\right)]$

Problem 3 (20 pts)

Let $X(e^{j\omega})$ denote the Fourier transform of the signal $x[n]$ shown in Figure 2. Perform the following calculations without explicitly calculating $X(e^{j\omega})$ using DTFT.

(a) Evaluate $X(e^{j0})$ and $X(e^{-j\pi})$

(b) Evaluate $\int_{-\pi}^{\pi} X(e^{j\omega})d\omega$

(c) Determine and sketch the signal whose Fourier transform is $Re\{X(e^{j\omega})\}$

(d) Evaluate: $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$ and $\int_{-\pi}^{\pi} |\frac{dX(e^{j\omega})}{d\omega}|^2 d\omega$

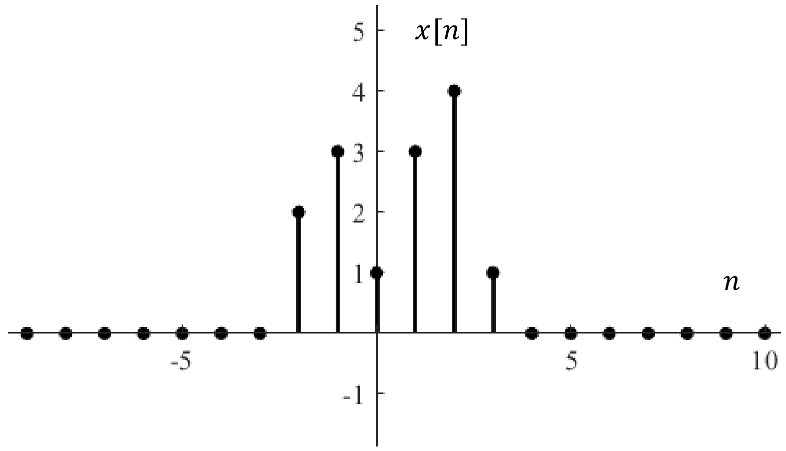

Figure 2: The discrete signal $x[n]$.

Problem 4 (25 pts)

Suppose we have an LTI system characterized by an impulse response

$$h[n] = \frac{\sin\left(\frac{4n\pi}{5}\right)}{n\pi}$$

(a) Find $H(e^{j\omega})$ using the duality between DTFT and continuous time Fourier series.

(b) Sketch the magnitude spectrum of the system's frequency response

(c) Sketch the spectrum of $X_1(e^{j\omega})$, $X_2(e^{j\omega})$, and $Y(e^{j\omega})$ given the input signal and the system block diagram.

$$x_1[n] = \cos\left(\frac{3\pi n}{4}\right)$$

$$x_2[n] = \frac{\sin\left(\frac{n\pi}{4}\right)}{n\pi}$$

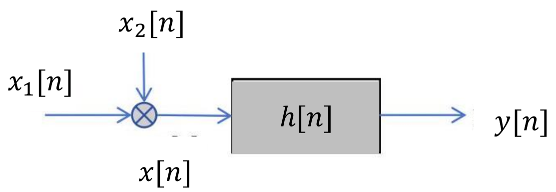

Figure 3: The system block diagram.

Problem 5 (10 pts)

A signal $x[n]$ is shown in Figure 4. Write $x[n]$ in the form of $y[n]$ defined as below. Find the DTFT expression of $x[n]$ using the time expansion property.

$$y[n] = \begin{cases} 1, & \text{for } n = 0,1,2,3,4 \\ 0, & \text{for other } n \end{cases}$$

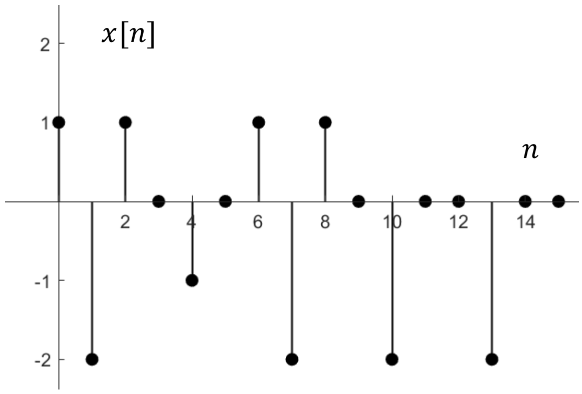

Figure 4: The discrete signal $x[n]$.

Problem 6 (20 pts)

A discrete-time, linear, time-invariant, and stable system with input denoted by $x[n]$ and output denoted by $y[n]$ is described by the difference equation:

$$y[n] + \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] + x[n-1] = x[n]$$

(a) Calculate the impulse response of the system $h[n]$. Is this system causal? Why or why not?

(b) Find $y[n]$ if the input is $x[n] = (\frac{1}{4})^n u[n]$